



**Technical School of
Telecommunications Engineering
and Systems**



ANTENNAS AND EMC

SOLUTION

**LABORATORY PRACTICE 4
APERTURE ANTENNAS**

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Table Of Contents

1. Introduction	3
2. Pyramidal-Horn Design.....	4
2.1. Theoretical Calculations	4
2.2. FEKO.....	8
2.2.1. Radiation Pattern, $G = 11.04$ dBi - Cartesian.....	8
2.2.2. Radiation Pattern, $G = 11.04$ dBi - Polar.....	9
2.2.3. E-Plane	10
2.2.4. H-Plane	10
2.2.5. Principal Planes	11
2.2.6. Symmetry Planes.....	11
2.3. Conclusions	12
3. Parabolic-Reflector Design.....	13
3.1. Theoretical Calculations	13
3.2. FEKO.....	14
3.2.1. H-Plane	16
3.2.2. E-Plane	16
3.2.3. Symmetry Planes.....	17
4. Radiation-Pattern Variation with Feed Displacement.....	18
4.1. Displacement $dx = 56.38$ mm.....	18
4.2. Displacement $dx = 112.76$ mm.....	18
4.3. Displacement $dx = 169.14$ mm.....	19
4.4. Comparison.....	20

1. Introduction

This laboratory exercise designs and simulates aperture antennas, focusing on pyramidal horns and parabolic reflectors modeled with FEKO. Digits from the assigned identification number set the exact operating frequency in the 9 GHz range and the required design parameters, including beamwidth and illumination levels.

The work has two parts. First, an optimum pyramidal horn is designed from the supplied universal curves and tables to meet gain and phase-error requirements. The horn is then used to illuminate a 1 m diameter parabolic reflector. A key step is locating the horn phase center at the reflector focus rather than placing the physical aperture center there. FEKO simulations are used to verify the resulting radiation pattern and aperture efficiency. Finally, the beam deflection produced by lateral feed displacement is studied.

$$DNI = 02757023J$$

$$f = 9000 + 200 \cdot 0 + 20 \cdot 2 + 2 \cdot 3 = 9046 \text{ MHz} = 9,046 \text{ GHz}$$

2. Pyramidal-Horn Design

2.1. Theoretical Calculations

Data:

$$a = 22,86 \text{ mm}$$

$$b = 10,16 \text{ mm}$$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{9,046 \times 10^9} = 3,3164 \text{ cm}$$

Given:

$$\theta = 45^\circ$$

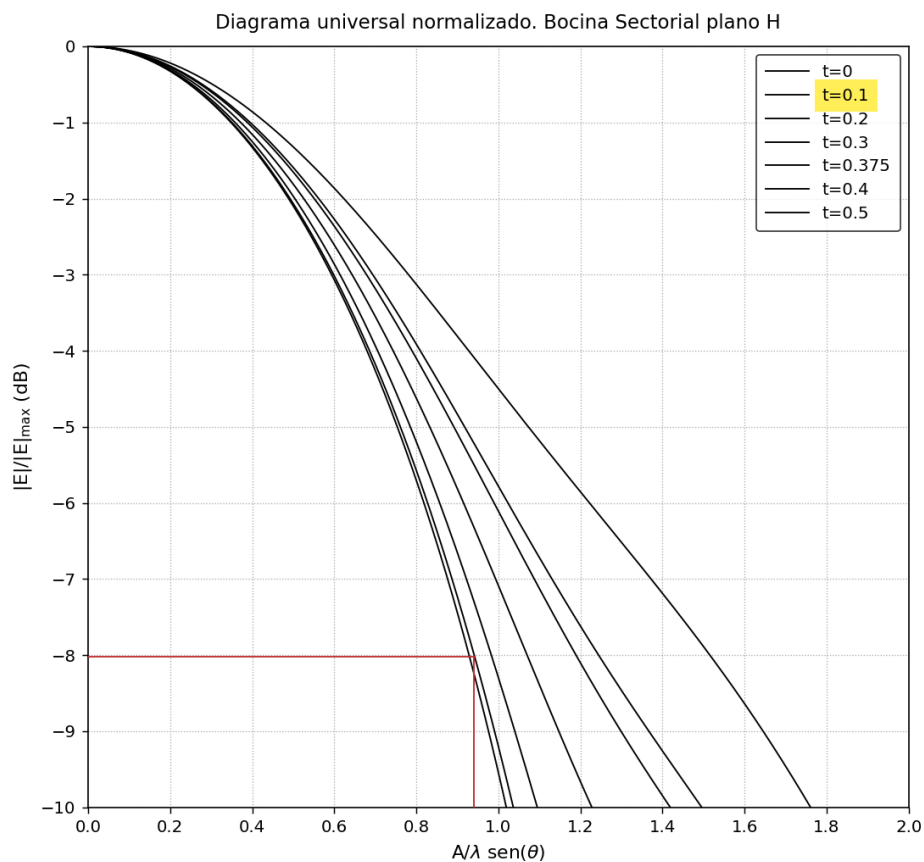
$$Caída = 9,4 \text{ dB}$$

The factor:

$$20 \log_{10} \frac{1 + \cos(\theta)}{2} = -1,38 \text{ dB}$$

We use the universal curves for a beamwidth of:

$$9,4 - 1,38 = 8,02 \text{ dB} \text{ y } t = 0,1$$



We obtain:

$$U_H = \frac{A}{\lambda} \times \sin(45^\circ) = 0,94$$

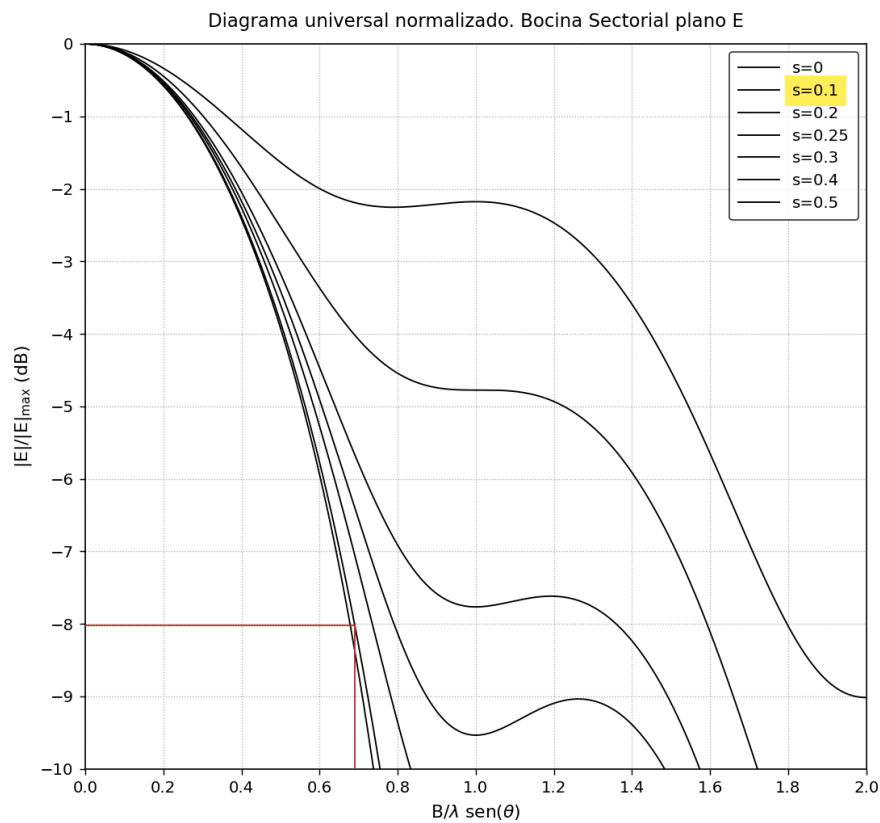
$$A = \frac{0,94 \times 3,3164}{\sin(45^\circ)} = 4,4087 \text{ cm}$$

$$R_1 = \frac{A^2}{8\lambda t} = \frac{4,4087^2}{8 \times 3,3164 \times 0,1} = 7,3259 \text{ cm}$$

$$R_H = R_1 \frac{A - a}{A} = 3,5272 \text{ cm}$$

To determine B, we use:

$$\Delta_{SemiAnchoDeHaz} 9,4dB = 45^\circ$$



We perform an iterative calculation.

Starting with:

$$B = A = 4,4087 \text{ cm}$$

For a realizable horn, the following condition must hold:

$$R_H = R_E = 3,5272 \text{ cm}$$

$$R_2 = R_E \frac{B}{B-b} = 3,5272 \times \frac{4,4087}{4,4087 - 1,016} = 4,5835 \text{ cm}$$

$$s = \frac{B^2}{R_2 8\lambda} = \frac{4,4087^2}{4,5835 \times 8 \times 3,3164} = 0,160$$

From the universal curves:

$$s = 0,160 \rightarrow U_E = 0,71 \rightarrow B = \frac{U_E \lambda}{\sin(45^\circ)} = 3,3300 \text{ cm}$$

Using this value of B, we recalculate:

$$R_2 = R_E \frac{B}{B-b} = 3,5272 \times \frac{3,3300}{3,3300 - 1,016} = 5,0760 \text{ cm}$$

$$s = \frac{B^2}{R_2 8\lambda} = \frac{3,3300^2}{5,0760 \times 8 \times 3,3164} = 0,082$$

$$s = 0,082 \rightarrow U_E = 0,69 \rightarrow B = 3,2362 \text{ cm}$$

Repeating the check once more, B no longer changes:

$$s = 0,077 \rightarrow U_E = 0,69 \rightarrow B = 3,2362 \text{ cm (converge)}$$

The final values are:

$$A = 44,1 \text{ mm}$$

$$B = 32,4 \text{ mm}$$

$$R_E = R_H = 35,3 \text{ mm}$$

$$s = 0,077 \approx 0,08$$

$$t = 0,1$$

The total efficiency is obtained from the following table:

s	Unif.	cos	cos ²	s	Unif.	cos	cos ²
0.05	0.04	0.02	0.01	0.55	4.93	2.09	1.16
0.10	0.15	0.07	0.04	0.60	5.91	2.44	1.36
0.15	0.34	0.16	0.09	0.65	6.96	2.82	1.57
0.20	0.62	0.29	0.16	0.70	8.04	3.20	1.79
0.25	0.97	0.45	0.25	0.75	9.08	3.58	2.01
0.30	1.40	0.65	0.36	0.80	9.98	3.95	2.23
0.35	1.92	0.88	0.49	0.85	10.60	4.31	2.46
0.40	2.54	1.14	0.64	0.90	10.87	4.65	2.69
0.45	3.24	1.43	0.80	0.95	10.80	4.97	2.91
0.50	4.04	1.75	0.97	1.00	10.50	5.25	3.13

$$s = 0,08 \rightarrow \eta_E^F = 0,11 \text{ dB}$$

$$s = 0,08 \rightarrow \eta_E^A = 0 \text{ dB}$$

$$t = 0,1 \rightarrow \eta_H^F = 0,07 \text{ dB}$$

$$t = 0,1 \rightarrow \eta_H^A = 0,91 \text{ dB}$$

Ideal aperture directivity:

$$D_0 = 10 \log_{10}[(4 \pi / \lambda^2) AB] = 12,13 \text{ dBi.}$$

$$\eta_{tot} = 0,91 + 0,11 + 0,07 = 1,09 \text{ dB}$$

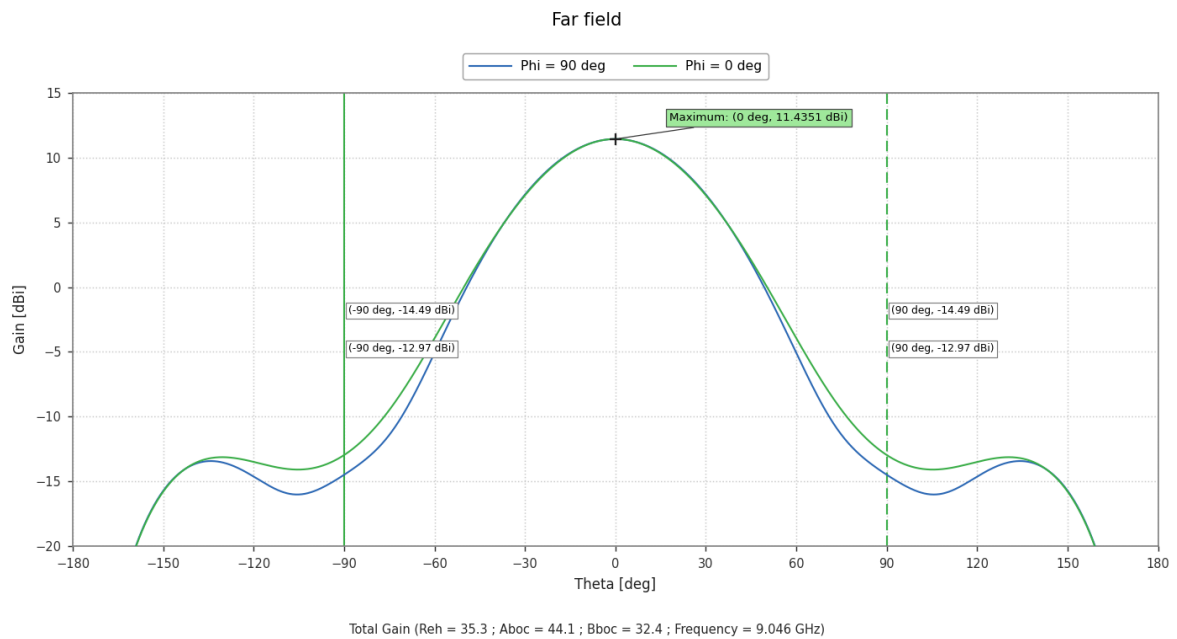
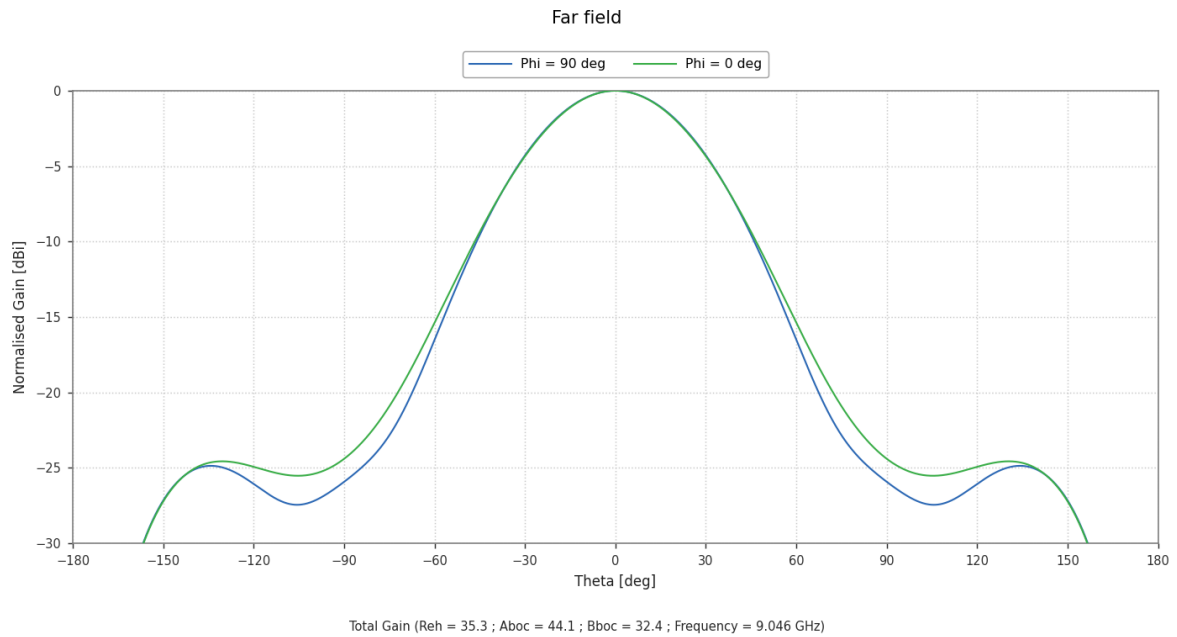
The directivity is calculated as:

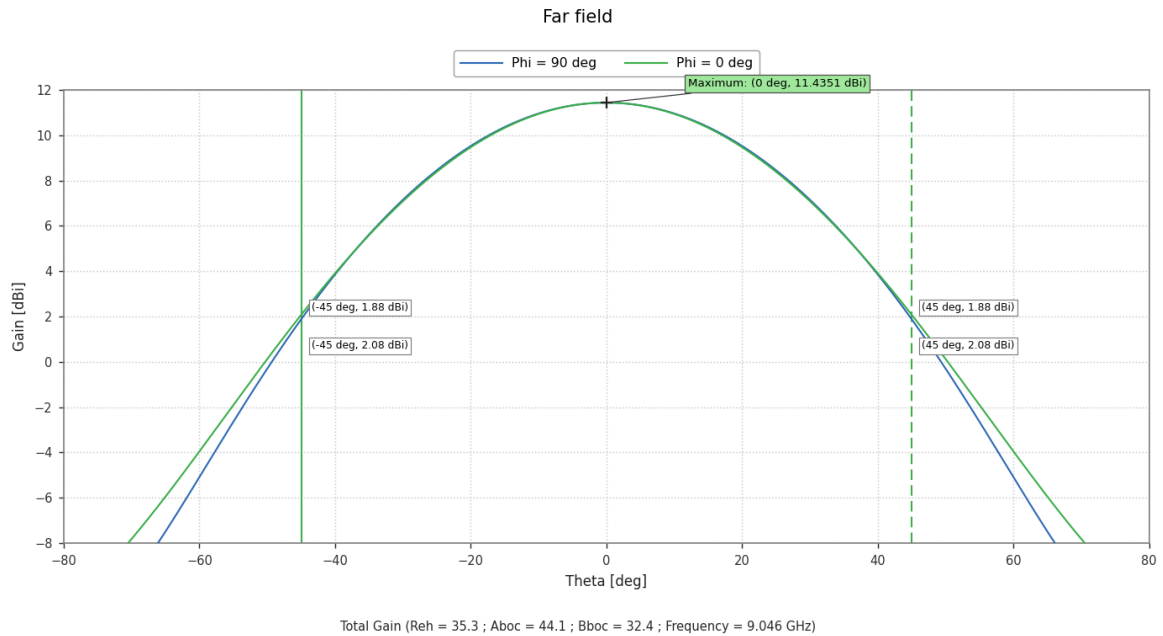
$$D_o = 10 \log \left(\frac{4 \pi}{\lambda^2} AB \right) = 12,13 \text{ dBi}$$

$$D_{total} = 12,13 \text{ dBi} - 1,09 \text{ dB} = 11,04 \text{ dBi}$$

2.2. FEKO

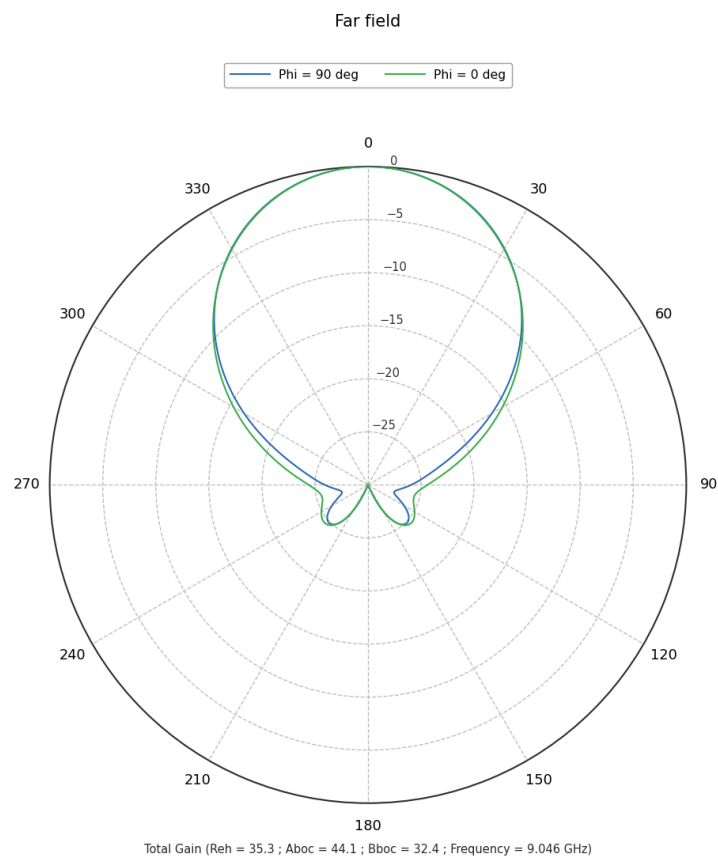
2.2.1. RADIATION PATTERN, $G = 11.04 \text{ dBi}$ - CARTESIAN



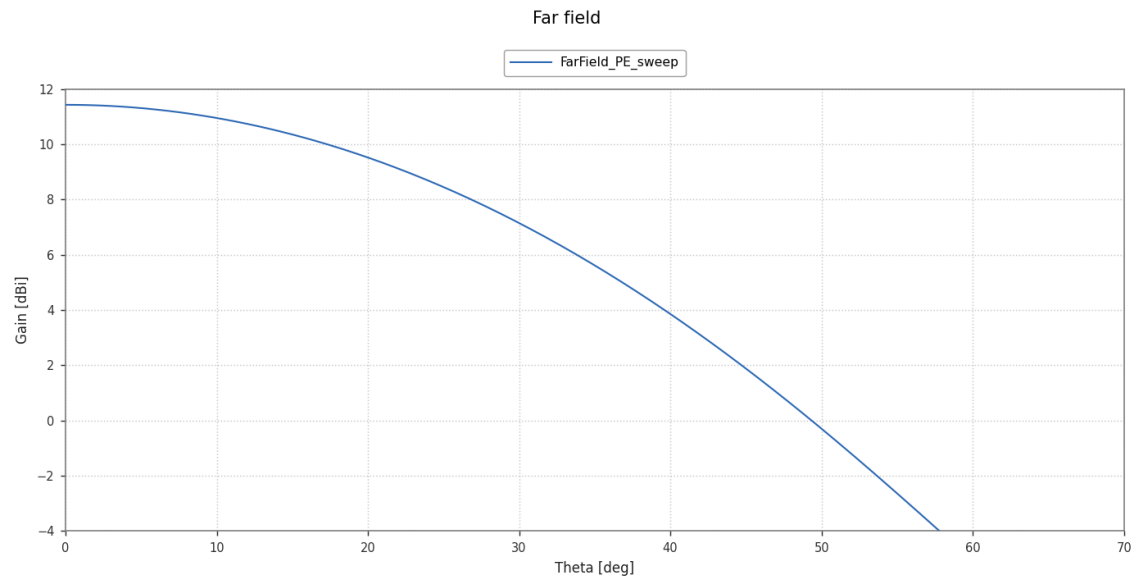


At the markers $\theta = \pm 45$ degrees, gain is 9.56 dB below the maximum in the E-plane and 9.36 dB below the maximum in the H-plane. Both values satisfy the specified 9.4 dB drop at a 45-degree half-beam angle.

2.2.2. RADIATION PATTERN, G = 11.04 dBi - POLAR

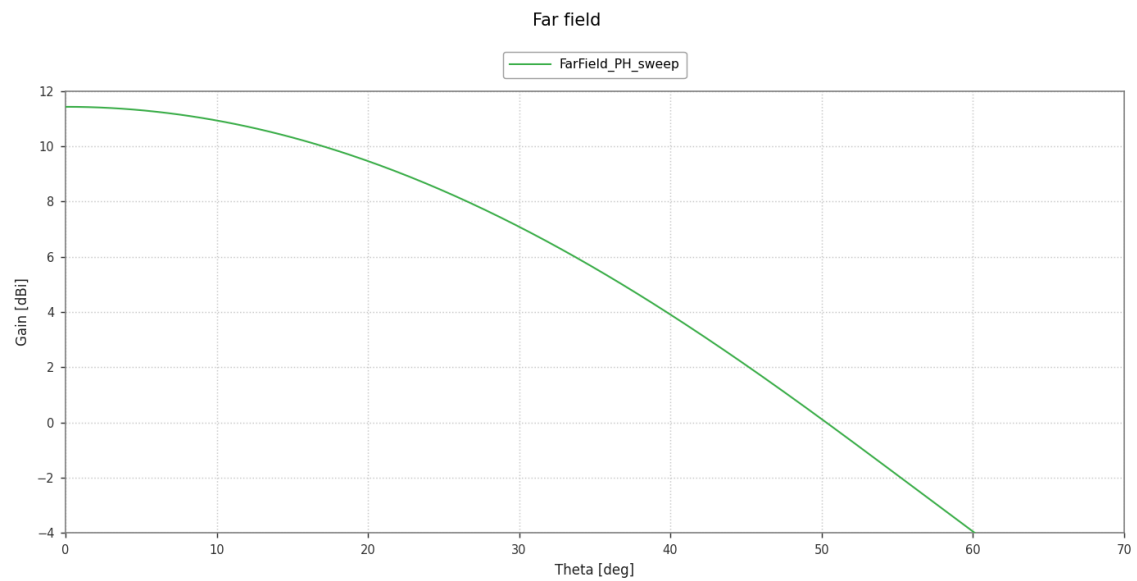


2.2.3. E-Plane



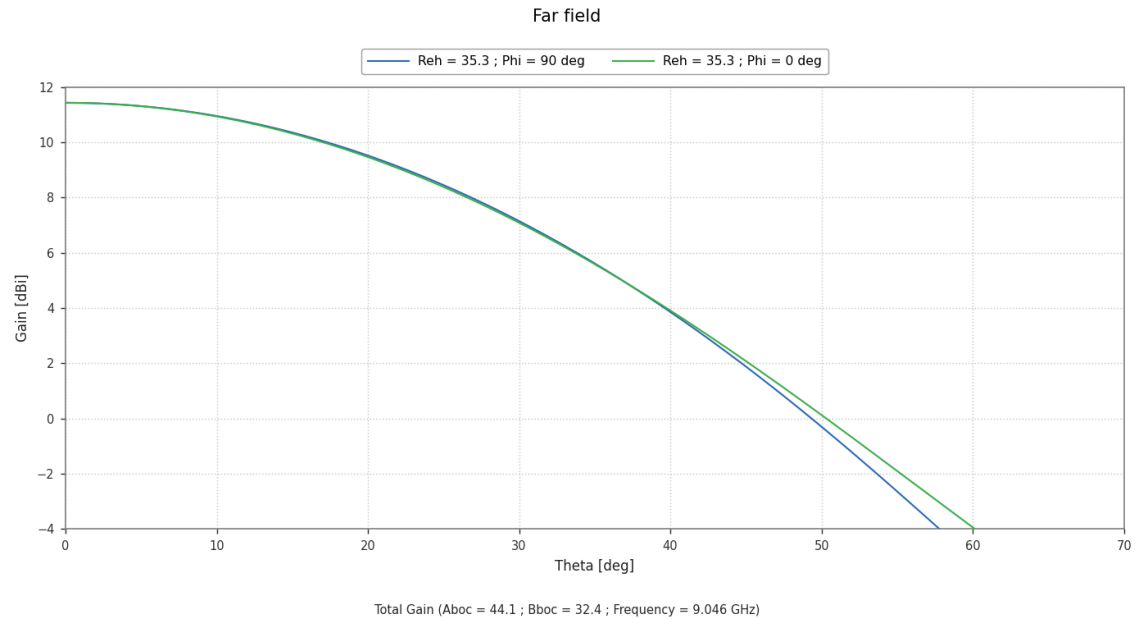
Total Gain [dBi] (Reh = 35.3 ; Aboc = 44.1 ; Bboc = 32.4 ; Frequency = 9.046 GHz; Phi = 90 deg)

2.2.4. H-Plane



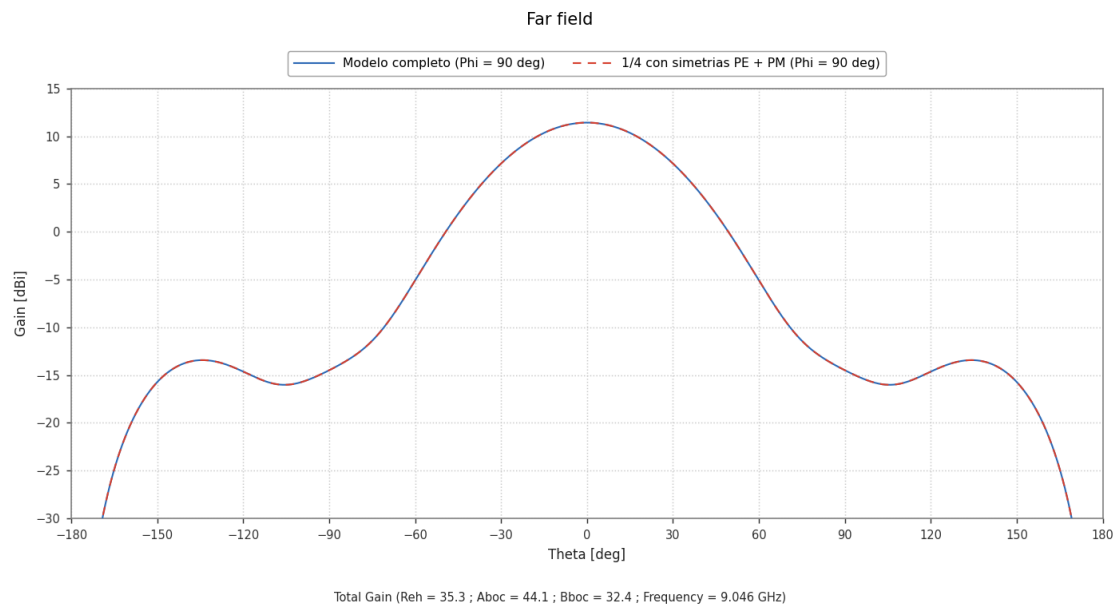
Total Gain [dBi] (Reh = 35.3 ; Aboc = 44.1 ; Bboc = 32.4 ; Frequency = 9.046 GHz; Phi = 0 deg)

2.2.5. Principal Planes



2.2.6. Symmetry Planes

The model has two symmetry planes. The YZ plane ($x = 0$), perpendicular to the magnetic field, is defined as a magnetic-symmetry plane. The XZ plane ($y = 0$), perpendicular to the electric field, is defined as an electric-symmetry plane. Enabling both planes in CADFEKO requires meshing only one quarter of the horn, while producing the same result as the complete model.



2.3. CONCLUSIONS

The theoretical design predicted 11.04 dBi, while FEKO produced a maximum of 11.44 dBi. The 0.40 dB difference is small and confirms that the universal-curve and table approximations are reliable for a standard pyramidal horn.

3. PARABOLIC-REFLECTOR DESIGN

3.1. Theoretical Calculations

Data:

$$D = 1000 \text{ mm}$$

$$\theta = 45^\circ$$

$$\theta = 2 \arctan \left(\frac{D}{4F} \right)$$

$$F = \frac{D}{4 \tan \left(\frac{\theta}{2} \right)} = \frac{1000}{4 \tan \left(\frac{45}{2} \right)} = 603,55 \text{ mm}$$

From theta = 2 arctan[D/(4F)]:					
s	L_h/R_h	L_e/R_e	s	L_h/R_h	L_e/R_e
0,00	0,0000	0,0000	0,28	0,258	0,572
0,04	0,0054	0,0110	0,32	0,334	0,755
0,08	0,0220	0,0450	0,36	0,418	
0,12	0,0480	0,1020	0,40	0,508	
0,16	0,0860	0,1820	0,44	0,605	
0,20	0,1340	0,2860	0,48	0,705	
0,24	0,1910	0,4160	0,52	0,808	

The horn phase-center corrections are obtained from the tabulated normalized distances. For $t = 0.1$ in the H-plane, interpolation gives $L_h/R_h = 0.0350$. For $s = 0.08$ in the E-plane, $L_e/R_e = 0.0450$.

$$\frac{L_h}{R_h} = 0,0350 \quad y \quad \frac{L_e}{R_e} = 0,0450$$

$$L_h = 0,0350 \times R_1 = 0,0350 \times 73,26 = 2,56 \text{ mm}$$

$$L_e = 0,0450 \times R_2 = 0,0450 \times 51,41 = 2,31 \text{ mm}$$

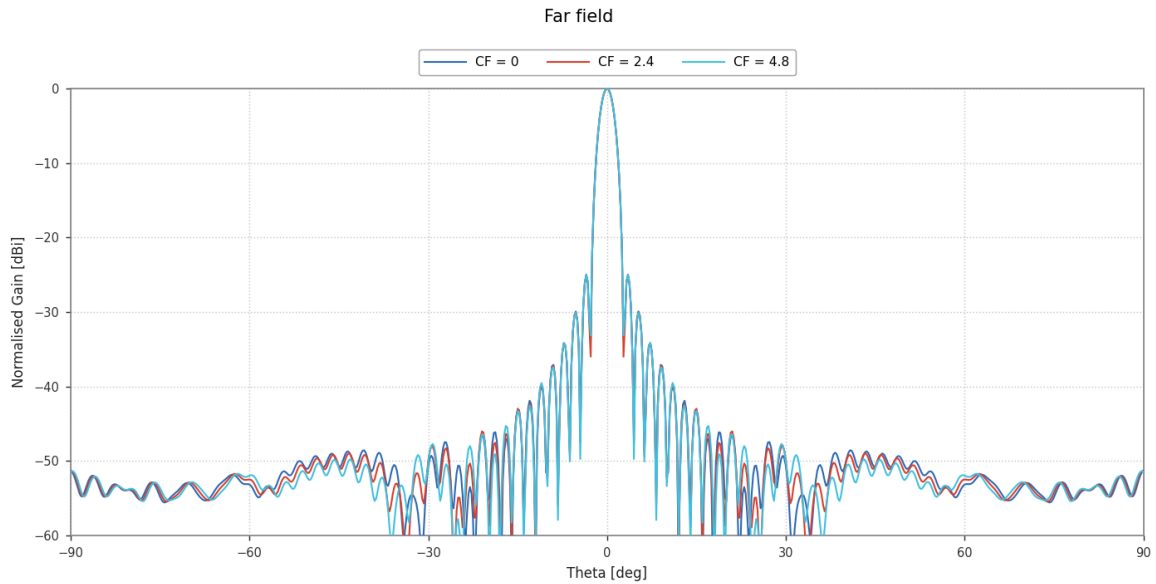
Thus, the phase center is approximately 2-3 mm behind the horn aperture. Using the average correction:

CF = $(L_h + L_e)/2$ approximately 2.4 mm.

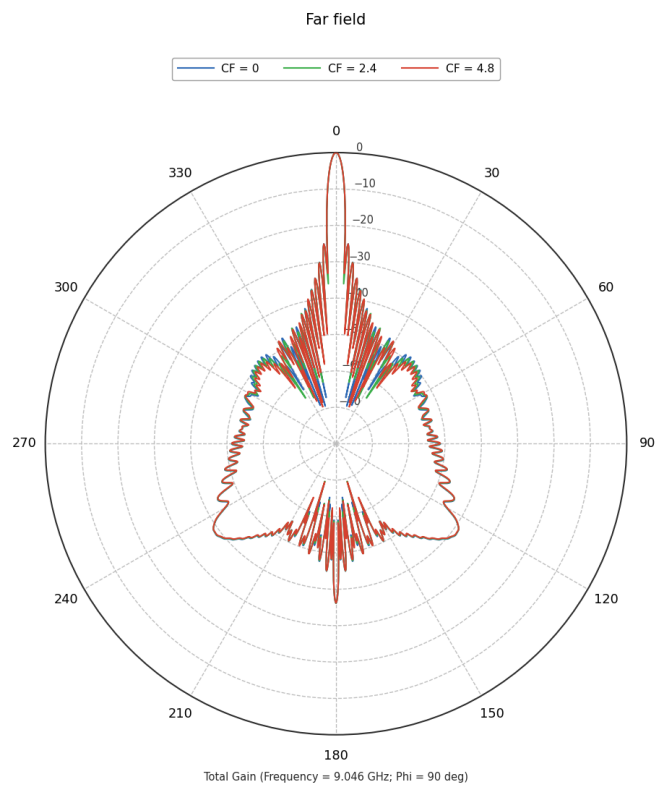
$$CF = \frac{L_h + L_e}{2} \approx 2,4 \text{ mm}$$

3.2. FEKO

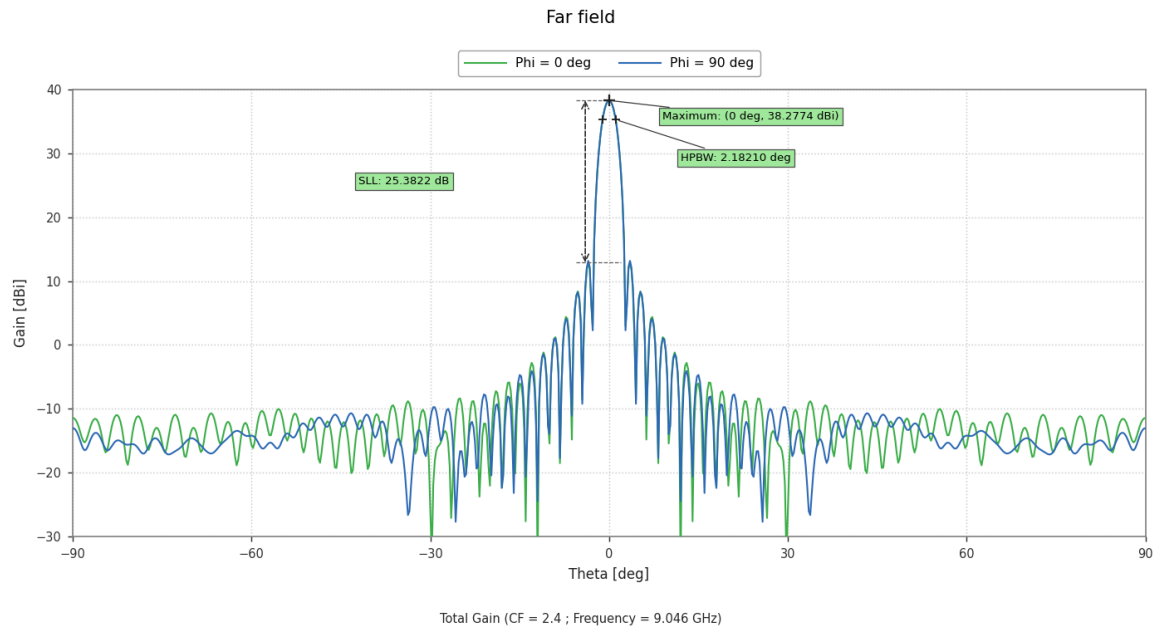
The horn and reflector are simulated with the calculated dimensions. To study the phase-center effect, three feed positions are compared:



Total Gain (Frequency = 9.046 GHz; Phi = 90 deg)



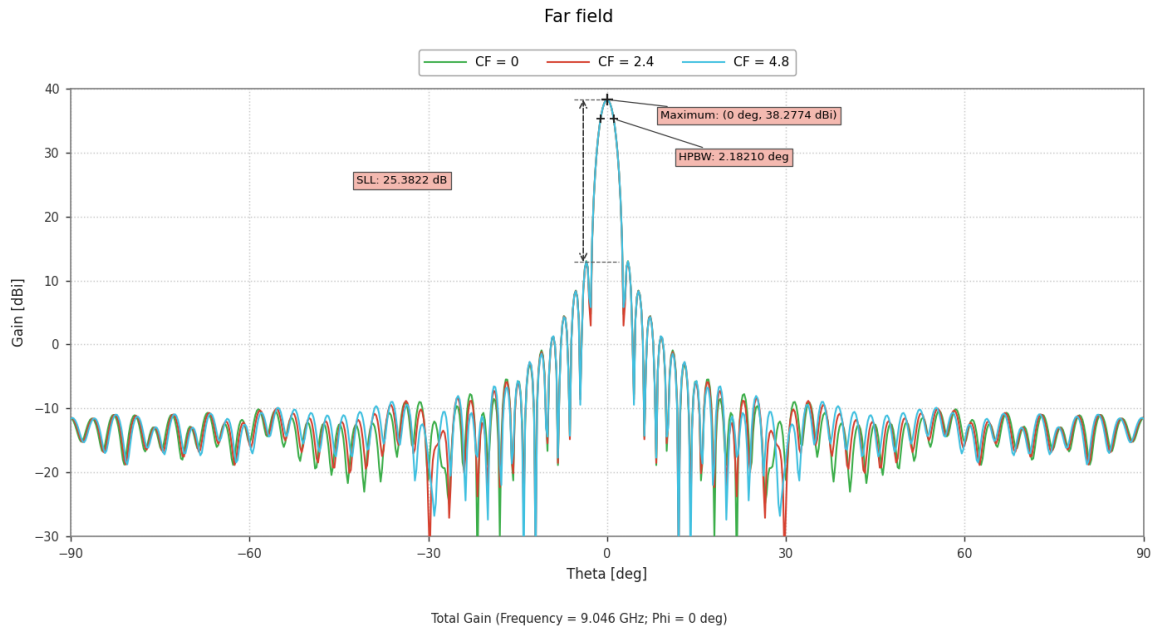
Comparando ambos planos:



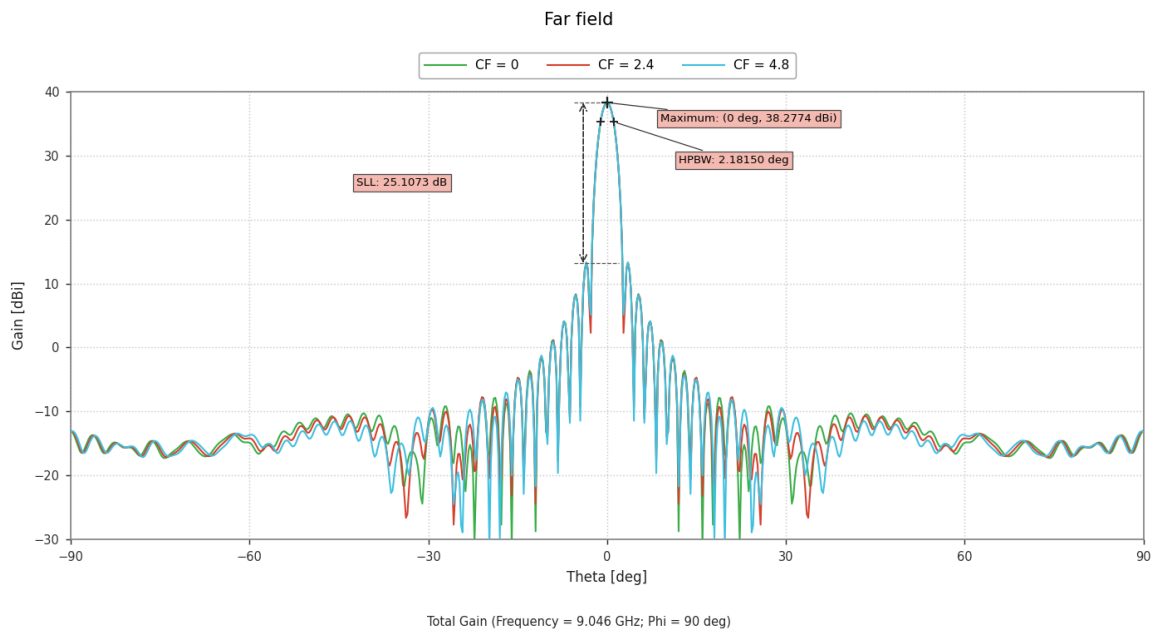
Placing the horn at its phase center produces a nearly ideal spherical wavefront that appears to originate at the reflector focus. This maximizes conversion to a plane wave, producing the highest gain, 38.28 dBi, and a symmetric radiation pattern.

If the horn were positioned using only its physical aperture center, residual phase error would slightly defocus the reflector, reduce aperture efficiency, and increase sidelobes. The corresponding gain is 38.27 dBi. The effect is small because the correction, about 0.07 lambda, is much smaller than the focal length, but the optimum remains the phase-center position.

3.2.1. H-Plane

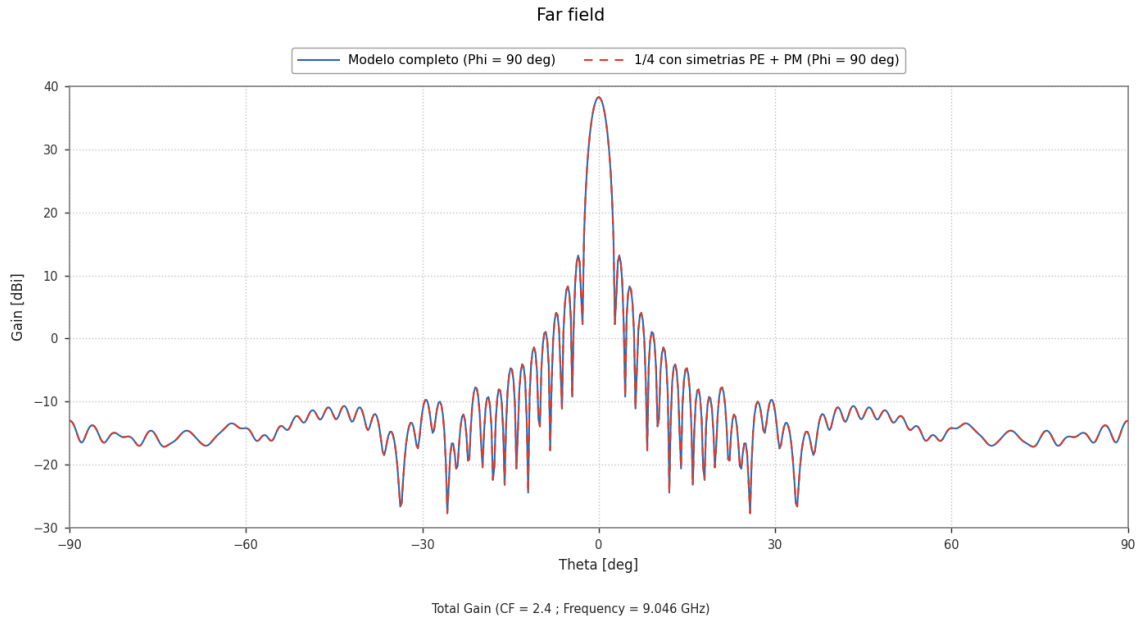


3.2.2. E-Plane



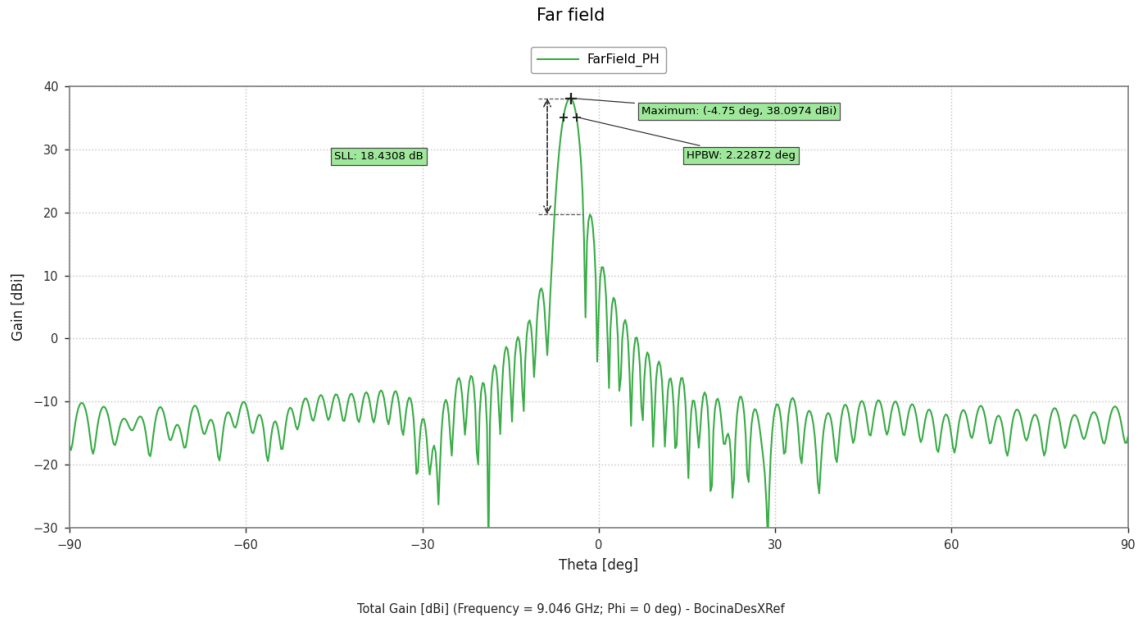
3.2.3. Symmetry Planes

With the feed centered, the horn-reflector assembly retains both horn symmetry planes: magnetic symmetry in YZ ($x = 0$) and electric symmetry in XZ ($y = 0$). Only one quarter of the reflector therefore needs to be simulated. The resulting radiation pattern is identical to the full-structure result.

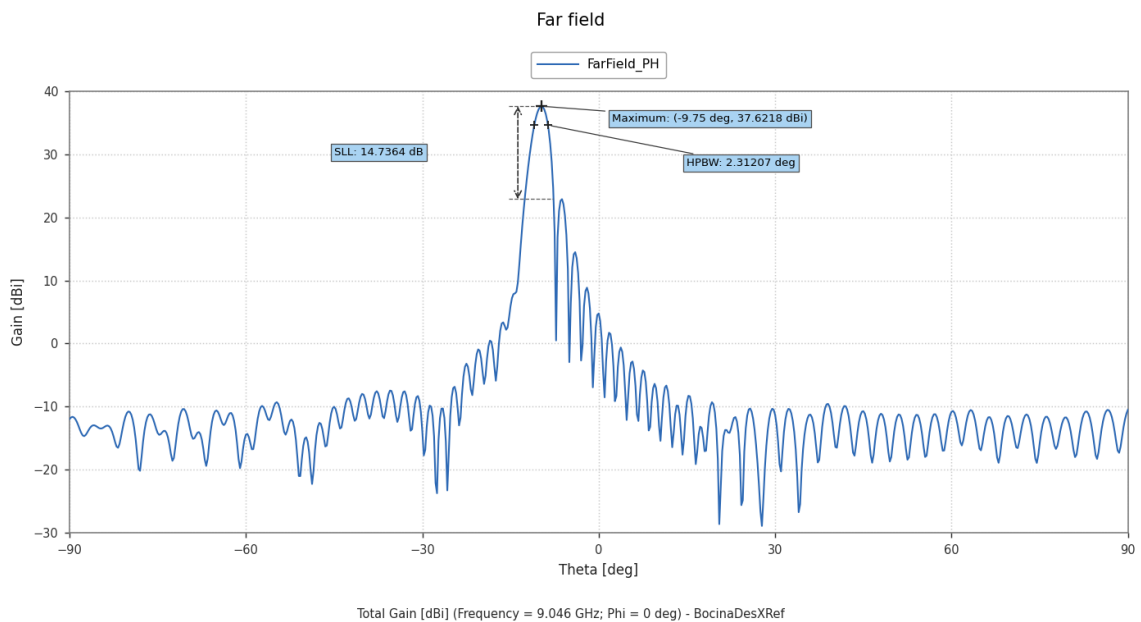


4. RADIATION-PATTERN VARIATION WITH FEED DISPLACEMENT

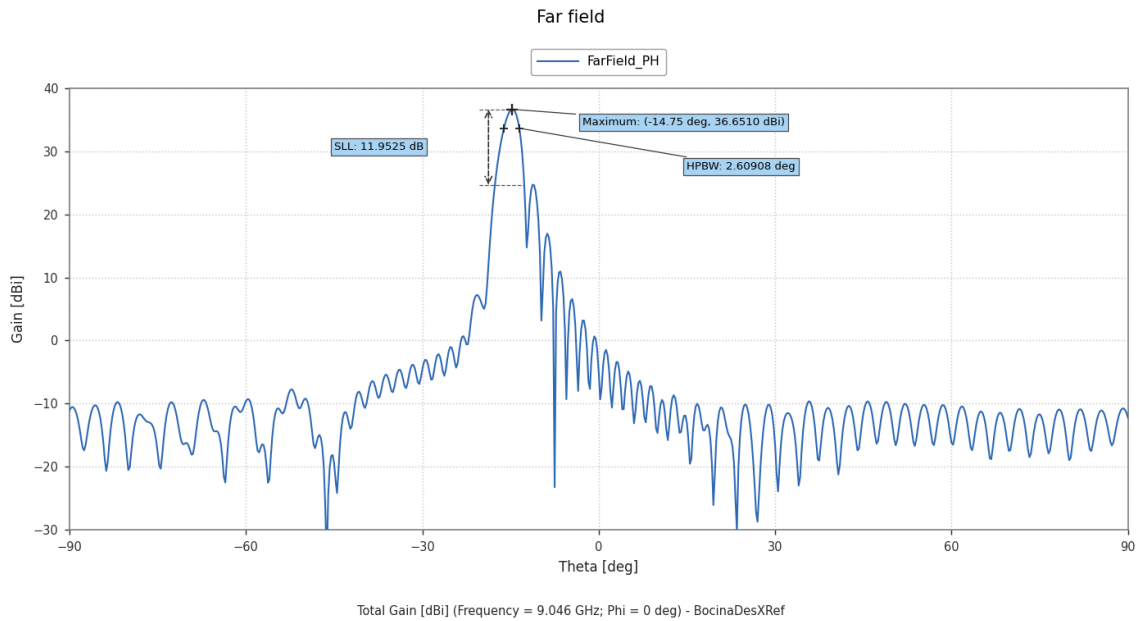
4.1. DISPLACEMENT $dx = 56.38$ mm



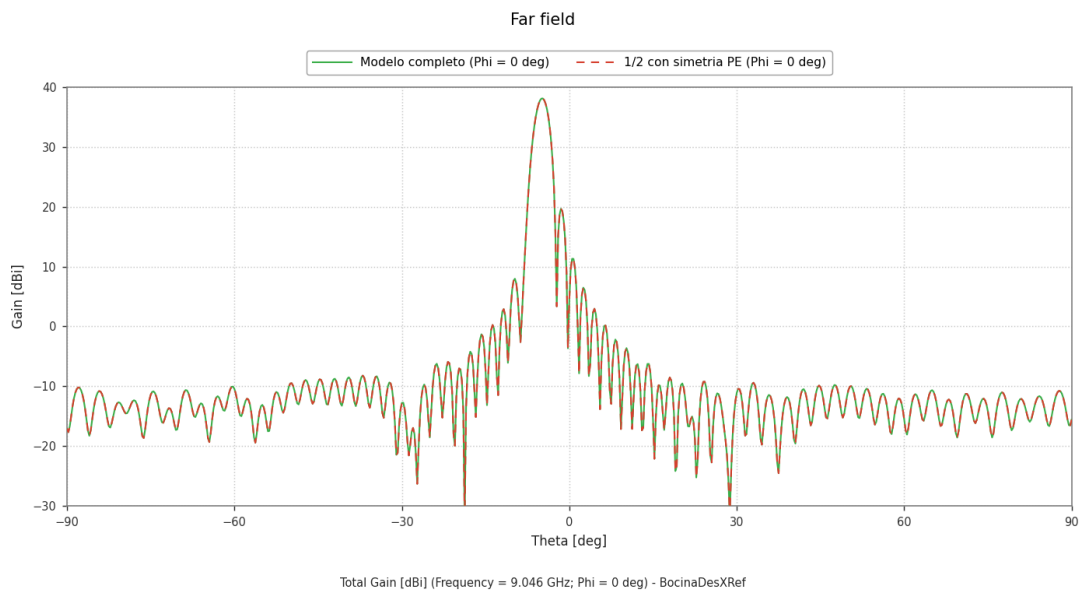
4.2. DISPLACEMENT $dx = 112.76$ mm



4.3. DISPLACEMENT $dx = 169.14$ mm



Lateral feed displacement along x breaks magnetic symmetry in the YZ plane. Electric symmetry in the XZ plane ($y = 0$) remains valid, so one half of the structure can still be simulated.



4.4. Comparison

		HPBW NLPLS Peak Position Gain		
0 d_x	2.18210 °	25.3822 dB	0°	38.2774 dB
1 d_x	2.22872 °	18.4308 dB	-4.75°	38.0974 dB
2 d_x	2.31207 °	14.7364 dB	-9.75°	37.6218 dB
3 d_x	2.60908 °	11.9525 dB	-14.75°	36.6510 dB

Feed offset | HPBW | Main-to-sidelobe level | Peak angle | Gain
 0 dx | 2.182 degrees | 25.382 dB | 0 degrees | 38.277 dB
 1 dx | 2.229 degrees | 18.431 dB | -4.75 degrees | 38.097 dB
 2 dx | 2.312 degrees | 14.736 dB | -9.75 degrees | 37.622 dB
 3 dx | 2.609 degrees | 11.953 dB | -14.75 degrees | 36.651 dB

As the feed moves in the positive x direction, the main beam tilts toward negative angles. Each displacement step moves the beam by about -4.75 degrees, close to the geometric estimate $\arctan(dx/F) = 5.3$ degrees after applying the reflector beam-deviation factor of about 0.9.

Gain decreases from about 38.3 dB on axis to 36.7 dB at the largest offset because illumination becomes less efficient and coma phase error increases. The largest deterioration is in sidelobe performance: the main-to-sidelobe level falls from more than 25 dB to about 12 dB. The coma lobe grows and partly merges with the main beam, increasing HPBW from 2.18 to 2.61 degrees.